# Luma Green connected platform specification

**Version 4  28 October 2026**

This is the business specification for the foundation release. The platform owner has delegated ordinary product decisions. Section 23 gives 40 selected decisions. Sections 26 to 30 add the detailed rules found in the full source review. The implementation team uses these rules unless the owner changes them.

The plan connects collection, stock, processing, buying, selling, service records, and source evidence. It uses one business and site access model. The current release is the web platform. On 28 October the owner paused Expo iOS and Android work. Existing mobile work is kept for later.

## What this document means

PLATFORM RULE describes the selected behavior. CURRENT RULE preserves an existing rule that still applies. A source note identifies where the business idea came from. Earlier source conflicts are resolved by the decisions in section 23. There is no remaining product questionnaire to complete.

The document defines what to build. It does not claim that every feature is already built, that a paid plan has been purchased, or that a provider or app store has accepted the release. A missing external connection results in an unavailable service, not simulated success.

## How the work connects

A person joins a business and a site. The site receives or offers material. A collection produces a receipt and stock movement. Stock can be sorted, processed, reserved, sold, and delivered. Each processing output retains its source. Each business order links the agreed terms, receipt, quality decision, invoice, and payment evidence. Reports read those records with the viewer access limits.

## Navigation

Sections 2 to 6 define scope, people, partners, and access. Sections 7 to 17 describe complete tasks. Sections 18 to 22 define screens, records, automatic work, fees, and acceptance. Section 23 records decisions. Sections 24 and 25 cover services and authority. Sections 26 to 30 give the source review supplement.

No programming examples, database instructions, terminal commands, or alternative software shopping lists are part of this business specification.

# 2 Purpose and first release

*Source: Excel Stakeholders C2 to V26; Chat Parts A D H and I; supplied project instructions*

**PLATFORM RULE** Luma.Green connects households, collectors, scrap shops, collection centres, yards, processors, recyclers, manufacturers, and buyers. It helps them arrange collection, record material, buy and sell, and keep evidence of what happened.

### What success means

A household gets a collection that turns up, clear weighing, and a receipt. A collector sees assigned work and payment records. A shop knows what it bought, holds, and sold. A processor can explain how input material became products and residual waste. A buyer can check the agreed quality and source history.

The business should be able to trace a material quantity through its recorded movements and processing. A receipt, payment record, or environmental claim must have supporting evidence.

### Working release boundary

**CURRENT RULE The web platform is the current release. Expo iOS and Android work is paused by the owner on 28 October. Keep existing mobile work for later. Electron desktop remains excluded. The owner permits replacing demonstration data and obsolete code without backward compatibility or migration work until this foundation version ships.**

**CURRENT RULE** The pilot location in the project instructions is Bengaluru. Carbon credit issuance and trading are later work. Real services must show an honest unavailable message when they are not configured.

**PLATFORM RULE** Build the connected collection and stock chain, business orders, processing records, and manufacturing source history. Use the same records for scoped reports and partner enquiries. Keep specialist execution unavailable until required evidence and provider connections are verified. Do not create a separate disconnected application for each role.

### Business boundaries to preserve

Keep material sale, collection service, processing service, and treatment service separate. A material owner may pay another business to process material without selling it. Waste with a treatment cost is a service purchase; do not hide that cost as a negative scrap price.

Implementation decisions: D01, D02. The selected answers are in section 23.

# 3 People who handle material

*Source: Excel Stakeholders rows 2 to 22 and 37 to 43; Chat Part A*

**PLATFORM RULE** A person can work for a business. A business can have more than one site and more than one activity. Being allowed to buy material does not automatically allow a business to process every material.

| Person or business | Main work and information needed |
|---|---|
| Household | Request collection or drop off; give location and photos; see own progress, receipt, and payment record. |
| Kabadiwala or scrap shop | Set own offers; arrange pickup; weigh and sort material; record purchases, stock, sales, and household payments. |
| Saathi or collection worker | Accept assigned work; collect and hand over material; record evidence and own earnings. Independent material selling is a separate proposed activity. |
| Dry waste centre or cooperative | Receive and sort dry waste; record weights and bales; sell sorted material; provide agreed collection reports. |
| Large yard or aggregator | Buy bulk stock; weigh, sort, bale, store, and dispatch; maintain lot and vehicle records. |
| Preprocessor | Shred, wash, bale, or prepare material. Record each run, output, quality result, and residual. |
| Recycler | Convert accepted material into usable output; keep source and processing evidence. |
| Intermediate manufacturer | Buy suitable input and produce an intermediate product, such as fibre or yarn. |
| Finished goods manufacturer | Buy input and make finished goods. Keep source evidence for any recycled content claim. |
| Institution or small shop | Arrange regular or on demand collection; manage vendors; receive weight and cost reports. |
| Factory or construction scrap seller | List stock or invite quotes; manage weighing, gate release, documents, and payment. |
| Brand or producer responsibility organisation | Procure material or services; track agreed obligations and eligible official evidence. |
| Specialist collector or processor | Handle approved electronic, wet, chemical, vehicle, or other specialist streams only within the separately approved scope. |

Implementation decisions: D03. The selected answers are in section 23.

# 4 Staff and people who review records

*Source: Excel Stakeholders columns D E and H rows 2 to 26 41 and 43; Chat internal permissions*

**PLATFORM RULE** Give access by work responsibility and record scope. A family relationship or job title alone is not enough to define permission.

| Staff or reviewer | Proposed responsibility and access limit |
|---|---|
| Business owner or manager | Manage the business team and its own work. Owners and authorized admins manage the permitted business approvals. |
| Purchasing or sales staff | Request and compare quotes, prepare orders, and manage supplier or customer communication. |
| Stores and weighing staff | Record received and dispatched quantities, storage location, and weighing evidence. |
| Production operator | Record input, work performed, output, and loss for assigned runs. |
| Quality staff | Record tests and results. A separate approved person may release or reject material. |
| Accounts staff | Prepare invoices and compare orders, receipts, and payments. Access follows the selected workspace roles. |
| Compliance staff | Maintain permission documents, due dates, and evidence. Do not grant an official permission on Luma’s behalf. |
| Household linked person | See or act on specifically shared household work. The sheet names relatives and house help; access follows the explicitly shared task. |
| NGO or worker union | Support workers, training, identity evidence, and sponsored programmes. No general right to private trade records. |
| City or ward official | Read the agreed reporting scope; add inspection remarks separately. Do not rewrite business trades. |
| Sponsor lender or auditor | Read explicitly shared evidence; record their own review. Loan or grant terms belong to that partner. |
| Luma support and operations | Handle support and permitted reviews. Current project rules separate member and admin access. |

Implementation decisions: D04, D05, D06. The selected answers are in section 23.

# 5 Proposed partners and wider services

*Source: Excel Stakeholders rows 27 to 36 and 39 to 51; Review Notes assumptions 6 7 and 9; Chat wider roles*

**PLATFORM RULE** The sources list the following partners. Their presence in this list does not prove a signed partnership, a working connection, or launch approval.

| Partner or service | Proposed purpose |
|---|---|
| Staff provider | Supply workers; keep deployment and agreed payroll evidence. |
| Equipment and scale supplier | Quote for equipment; deliver and service it; keep scale verification evidence. |
| Transport business | Assign drivers and vehicles; record collection, delivery, and freight charges. |
| Water treatment and environmental engineer | Survey a site; propose, install, or service treatment equipment. |
| Boiler items and tanker supplier | Supply agreed items or tanker services. The customer states the item and service details in each enquiry. |
| Accountant | Prepare and review invoices, tax information, and filing records. |
| Solar supplier | Survey, quote, install, and service solar equipment. Proposed later service. |
| Attendance service | Record attendance or agreed work travel with privacy limits. |
| Payment partner | Process supported payments and return payment status and settlement records. |
| Rewards merchant | Redeem approved coupons or rewards under a funded agreement. |
| Insurance or welfare partner | Handle eligibility, enrolment, and claims under its own rules. |
| Society app or facility manager | Pass approved pickup requests and coordinate entry to the premises. |
| Community volunteer or school | Run campaigns and help people make a first request. |
| Voice and local language service | Support spoken prompts, calls, or assisted requests. |
| Rural body or self help group | Plan village collection, sort stock, sell to buyers, and record group proceeds. |
| Certificate or carbon market partner | Handle approved official evidence or credits. This is separate from ordinary material stock. |
| Lab specialist handler or repair buyer | Test material, handle residuals, or assess reuse, if the relevant service is approved. |

Implementation decisions: D07. The selected answers are in section 23.

# 6 Accounts sign in and access

*Source: Excel Stakeholders G and H; Chat organisation and facility model; supplied project instructions*

### Joining and sign in

**CURRENT RULE** Phone users sign in without a password. The configured admin phone also needs a code from an authenticator app. Keep all required sign in factors. Do not offer a pretend sign in when the delivery service is unavailable.

**CURRENT RULE** Keep the current Surepass business registration check and explicit business name and address confirmation. Retain optional business PAN to GST discovery for the current eligible registered business flows. Household, Saathi, and admin accounts do not enter that business discovery flow. Keep Saathi identity and selfie evidence. Tax registration does not prove operating permission.

### Access after sign in

**PLATFORM RULE** A person sees their own account and the business, site, or household records shared with them. A driver sees assigned trips. A worker sees assigned collections. A financial reviewer sees the records needed for that review. An auditor receives a limited view rather than control of the underlying record.

**CURRENT RULE** Record access also controls private photos and documents. Removing access must prevent later access to those files. Support and admin screens must not let a member act as an admin.

### Recovery and leaving a group

**PLATFORM RULE** Losing a phone, changing a phone, leaving a business, and sharing a phone are different cases. Each needs a safe recovery or removal process. Keep the person’s identity and required security checks when a phone is linked to an existing account.

**Checks to demonstrate** Another business cannot open a private record or photo. A removed member loses access. An admin must complete all required sign in checks. A failed sign in message preserves no unauthorized access.

Implementation decisions: D08, D09, D10. The selected answers are in section 23.

# 7 TASK 01 Household pickup or drop off

*Source: Excel Stakeholders rows 2 to 8; Chat household use case; supplied project instructions*

**PLATFORM RULE** The household wants a reliable handover and a clear receipt. It can request pickup or select drop off. The phone must be verified before the request is accepted.

**CURRENT RULE** Collect photos, address or location, pickup or drop off choice, time, name, and verified phone. Do not require household material estimates or weights. The shop records actual materials and weight later. Keep entered values when the person moves between steps.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Choose Recycle and pickup or drop off. | Show the information needed for that choice. |
| 2 | Add photos location time and contact details. | Show missing information and preserve completed answers. |
| 3 | Review and submit. | Save one request and show a clear confirmation. |
| 4 | Check progress and hand over material. | Show the accepted arrangement and later receipt. |
| 5 | Review the recorded weight and payment. | Show evidence and a way to raise a complaint. |

### Information and evidence

Keep the request, contact details, selected time, photos, assigned business when known, progress history, receipt, and household payment acknowledgement. Photos stay private. Do not present a photo estimate as the final weighing result.

### Exceptions and recovery

If a photo fails to upload, explain how to retry. If location permission is refused, provide a usable address entry. If submission is repeated, check for an existing request. If no collector accepts, show that the request is still unassigned; do not show a confirmed visit.

**Acceptance check** A household can submit without material names or weights. Repeated submission makes one request. Only permitted people can read the photos. A refused location permission does not prevent manual address entry.

Implementation decisions: D11, D12, D13. The selected answers are in section 23.

# 8 TASK 02 Collection work and handover

*Source: Excel Stakeholders rows 9 to 16 and 40 to 42; Chat Saathi and household use cases*

PLATFORM RULE A permitted business assigns work. Household pickup handovers link the assigned shop job and visit. The worker sees the job details and agreed fixed fee. The handover does not grant household contacts, address, photos or material rate access.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Worker accepts or declines assigned work. | Show the response to the person who assigned it. |
| 2 | Worker travels to the agreed place. | Show only the necessary active work details. |
| 3 | Worker records the measured collected total for the assigned visit. | Keep exact grams and an evidence reference. The handover gives no household contact or payment access. |
| 4 | The shop completes the household receipt. | Record measured material lines and add stock once through the existing collection record. |
| 5 | Worker delivers to the shop or centre. | Compare collected and received quantities and keep the handover record. |
| 6 | Payer records household payment or worker pay. | Keep separate material and work payment records. |

### Money and responsibility

A worker fee is fixed per job and agreed before acceptance. The job does not make the worker the material owner. Independent worker selling and worker owned stock are outside this version. Keep material payment separate from worker pay. Collection evidence cannot change an agreed rate or payment.

### Failure paths

If the household is absent, record a failed visit and the reason. If material is unsafe or different from the request, stop and ask the responsible person. If collected and delivered weights differ, retain both records and raise a review. Do not silently reduce the earlier collection.

Acceptance check A worker can open assigned work but not unrelated household records. The shop receipt adds stock once; handover evidence adds none. Material payment and worker wages remain distinguishable.

Implementation decisions: D14, D15. The selected answers are in section 23.

# 9 TASK 03 Rates stock and correction

*Source: Excel Stakeholders J to O rows 9 to 20; Chat pricing and inventory controls*

**PLATFORM RULE** A shop, yard, or processor maintains rates and stock by material and grade. Each rate states its unit, validity, location, and relevant delivery terms. Keep buyer bids, seller asking prices, observed trades, and indicative guidance visibly different.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Authorized person enters a rate or offer. | Keep the previous rate, new rate, date, and person responsible. |
| 2 | Operator receives and weighs material. | Create stock linked to its source and evidence. |
| 3 | Operator sorts stores or bales material. | Keep each changed form and the source quantities. |
| 4 | Operator reserves or sells available stock. | Prevent the same quantity being promised or consumed twice. |
| 5 | Operator completes the day closing or adjustment. | Compare opening stock, receipts, use, sales, and closing stock; preserve corrections. |

### Stock value

The sheet proposes average purchase cost, landed cost, current selling value, stock age, and low stock alerts. These are different views. A change in market rate must not change an old agreed price or recorded payment. Internal stock movements and processing are not automatically marketplace sales.

**CURRENT RULE** Money and mass use exact amounts. Preserve paise and grams. Every state change must have an audit record. Do not permanently erase a completed financial or material history through an ordinary delete action.

**Acceptance check** The same stock cannot be sold twice. An adjustment retains the earlier value and reason. An unavailable live rate is shown as unavailable.

Implementation decisions: D16, D17, D18. The selected answers are in section 23.

# 10 TASK 04 Find suppliers agree terms and order

*Source: Excel Stakeholders rows 16 to 22 and 38; Chat Source to Pay and Procure to Pay sections*

**PLATFORM RULE** Businesses may discover suppliers and compare quotes before buying. A routine repeat purchase can use agreed terms. Household collections use the simpler task in section 7.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Buyer states material grade quantity and delivery need. | Find suitable offers or invite quotes. |
| 2 | Supplier states price availability and terms. | Show like for like quotes, including delivery and other charges. |
| 3 | Buyer checks supplier capability and any required evidence. | Show missing or expired evidence for the proposed activity. |
| 4 | Authorized people agree a quote or contract. | Keep the accepted version, quality conditions, price basis, and approval. |
| 5 | Buyer places the order and seller accepts. | Reserve the agreed stock or capacity and show order progress. |
| 6 | Both sides arrange receipt and invoicing. | Link the order to delivery, inspection, invoice, and payment evidence. |

### Terms that must be visible

Identify the seller, buyer, sites, material, grade, quantity basis, rate, rate validity, freight payer, delivery point, acceptance test, deductions, payment timing, and cancellation terms. Show separate material, transport, testing, and platform charges.

### Four kinds of order

Material purchase transfers ownership as agreed. Collection service pays for collection work. Processing service pays another business to process owner held material. Treatment service pays an approved handler. Do not force all four into a scrap sale.

**Acceptance check** A business sees all applicable charges before acceptance. The accepted terms remain available after rates change. Required eligibility cannot be bypassed by a higher price.

Implementation decisions: D19, D20. The selected answers are in section 23.

# 11 TASK 05 Payment invoice refund and dispute

*Source: Excel Stakeholders P to S; Review Notes assumption 2; Chat settlement and revenue sections; supplied project instructions*

**CURRENT RULE** A kabadiwala pays a household directly and Luma records that separate flow. Business checkout uses the selected payment partner only when live use and the required policy are explicitly enabled. Do not mark a trade paid through a manual override.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Parties review agreed amounts and payment terms. | Show material value, charges, tax, and net amount separately. |
| 2 | Payer uses the supported payment method. | Show pending, confirmed, failed, or unknown payment honestly. |
| 3 | Operator carries out permitted dispatch and receipt. | Enforce the approved payment and delivery conditions. |
| 4 | Accounts person compares order receipt and invoice. | Flag differences and keep the supporting references. |
| 5 | Authorized person handles a complaint or permitted refund. | Keep the decision and verified refund or settlement evidence. |

### Failures and refunds

A lost connection is not proof of payment failure. Check payment status before asking for another payment. Repeated provider notices must not pay twice. Keep disputed weight and quality records. The current business refund process does not automatically restore stock or cancel the trade after a confirmed refund.

**Acceptance check** One payment confirmation produces one result. A failed or unknown payment cannot authorize an unsupported stock release. A refund retains its reason, approver, amount, and provider evidence.

Implementation decisions: D21, D22, D23. The selected answers are in section 23.

# 12 TASK 06 Process material and record outputs

*Source: Excel Stakeholders rows 17 to 20; Chat material model PET reference and processing use cases*

**PLATFORM RULE** A plant receives a defined input and records the work performed. One run may create several useful outputs and residual waste. The same business may perform more than one step; the material does not have to pass through every named business type.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Plant manager selects available input lots and work. | Show source, ownership, quantity, and required quality or permission checks. |
| 2 | Operator records the input actually used. | Deduct that quantity once from available input. |
| 3 | Operator records processing and every output. | Create linked output records for products, side streams, rejected material, and residuals. |
| 4 | Quality staff record required checks. | Keep material on hold until the permitted release decision. |
| 5 | Responsible person explains quantity differences. | Compare input with outputs, residuals, recorded additions, and explained losses. |
| 6 | Authorized person closes the run. | Preserve evidence and release only the approved output. |

### A plain material example

A PET bottle load can be sorted, baled, washed, and turned into flakes. Separate caps, labels, fines, wastewater, and other residuals must be recorded when present. Later plants may make fibre, yarn, fabric, and garments. Actual routes and suitable outputs depend on the plant and material.

The chat gives different technical and regulatory claims about these processes. Do not treat washing alone as proof that an output is suitable for food contact. Do not claim a quality property without a relevant approved test or evidence.

### Processing another owner’s material

A yard may keep ownership while a plant washes its material for a fee. Record the material owner, the plant holding it, quantities sent and returned, each output, and the service invoice. It is not an automatic material sale.

**Acceptance check** No input is consumed twice. Every output links to its source. Unexplained quantity differences remain open. Material ownership stays clear during paid processing.

Implementation decisions: D24. The selected answers are in section 23.

# 13 TASK 07 Dispatch receipt quality and returns

*Source: Excel Stakeholders rows 16 to 20 29 and 38; Chat delivery variance and quality failure examples*

**PLATFORM RULE** Seller dispatch, physical delivery, buyer inspection, and final acceptance are separate events. Keep both parties' weighing evidence and quality results.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Seller prepares the approved load. | Check available stock, agreed documents, vehicle, and destination. |
| 2 | Authorized person confirms dispatch. | Keep dispatch weight, time, load identity, and evidence. |
| 3 | Driver carries and hands over the load. | Record trip progress and delivery evidence within the agreed location limits. |
| 4 | Buyer records received weight and condition. | Keep received and dispatched figures without replacing either. |
| 5 | Quality reviewer accepts holds or rejects material. | Record quantity, test, reason, and the responsible decision. |
| 6 | Responsible parties resolve any return or adjustment. | Reconcile stock, invoice, payment, and any linked claim evidence. |

### A quantity difference example

The chat describes a load dispatched at 10,000 kilograms and received at 9,820 kilograms. Keep both measurements. Compare the agreed tolerance. Separate transit difference from quality rejection. The example does not set the permitted tolerance for every order.

### Exceptions

For a missing delivery, seal problem, wrong material, or quality failure, keep the affected stock and payment in the status required by the approved policy. Record evidence, responsible person, next action, and deadline. Do not silently replace an old weight or automatically route a specialist rejected load to an ordinary collector.

**Acceptance check** A quality hold prevents unauthorized sale or use. Every rejected quantity has a recorded disposition. An accepted return cannot create stock twice or leave an incorrect environmental claim active.

Implementation decisions: D25, D20, D22. The selected answers are in section 23.

# 14 TASK 08 Institutions municipal and rural collection

*Source: Excel Stakeholders rows 15 22 37 40 41 42 and 49; Chat programme reporting*

**PLATFORM RULE** An apartment committee, office, hotel, shop, factory, or local body can arrange collection and receive reports for its own site or agreed area. The amount of work differs from a single household request.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Site manager selects approved waste streams and vendors. | Keep service agreement, site, responsible contacts, and scope. |
| 2 | Manager books a one time or repeating service. | Show planned collection dates and vendor response. |
| 3 | Collector records the visit and handover. | Keep quantities by stream, missed visits, and receiver evidence. |
| 4 | Site representative confirms the record. | Show exceptions and relevant payment or service charges. |
| 5 | Authorized reviewer requests a report. | Show the agreed totals by site, period, stream, or area. |

### Municipal and rural work

The sheet proposes route coverage, vehicle trips, missed collection complaints, centre handovers, and ward reports. It also proposes rural self help group collection, group sales, wages, maintenance funds, and reinvestment. These are proposed operating models, not proof of a municipal agreement.

Separate a material sale from a paid collection service. Record any city contract payment for reference only unless a supported payment arrangement is approved. Do not turn a quantity report into a promise of legal compliance.

**Acceptance check** A site sees only its own permitted records. A schedule change does not alter a past completed visit. Repeated visits remain separate records. A ward report does not expose unrelated household contact details.

Implementation decisions: D01, D26, D27. The selected answers are in section 23.

# 15 TASK 09 Specialist waste and treatment

*Source: Chat compliance and later consignment sections; Excel Stakeholders rows 38 39 and 48*

**PLATFORM RULE** The sources propose specialist electronic waste, batteries, tyres, used oil, chemical and pharmaceutical residues, end of life vehicles, and treatment or disposal services. These need separate business and safety rules before they become available.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Generator identifies the stream and supporting evidence. | Mark uncertain classification for review instead of guessing. |
| 2 | Responsible reviewer checks the proposed route. | Show required source, transporter, receiver, capacity, and permission evidence. |
| 3 | Authorized parties agree handling service and charges. | Record treatment and transport charges separately from material value. |
| 4 | Responsible person prepares applicable movement records. | Keep the approved document version and any official acknowledgement. |
| 5 | Receiver records arrival acceptance and treatment. | Keep quantity differences, treatment evidence, and final outcome. |
| 6 | Reviewer reconciles the movement. | Close only when the required evidence and exceptions are resolved. |

### Source limits that need attention

The chat refers to other industry, pharmaceutical, and regulatory files that are not among the two supplied files. Their reported contents are leads for review, not a verified material catalogue. It gives conflicting industry totals, form numbers, consent periods, and broad routing claims.

A tax registration check does not prove permission to operate a treatment plant. A collection receipt does not prove recycling or final disposal. A proposed sale price does not prove that a waste may be sold or used for the suggested purpose.

**Acceptance check** An unknown or ineligible stream cannot be routed as an ordinary household collection. Missing required evidence blocks the applicable movement. The person sees the reason and the next safe action.

Implementation decisions: D28, D23. The selected answers are in section 23.

# 16 TASK 10 Source evidence brands and environmental claims

*Source: Excel Stakeholders rows 18 to 21 25 30 31 and 43; Chat claims and manufacturing sections; project instructions*

**PLATFORM RULE** A buyer, brand, sponsor, or auditor may need evidence of source, collection, processing, or recycled content. Different claims require different evidence and approval. One material movement does not automatically create every kind of certificate.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Responsible person states the claim or reporting need. | Identify the related product, period, material, scheme, and required evidence. |
| 2 | Businesses supply permitted source and processing records. | Link the evidence without exposing unrelated private records. |
| 3 | Reviewer checks eligibility and quantity. | Keep missing evidence, rejected evidence, and allocated quantities visible. |
| 4 | Authorized party obtains official approval where required. | Record the official reference and outcome; label drafts clearly. |
| 5 | Permitted viewer reads the approved evidence report. | Show scope, source limits, and later corrections or withdrawal. |

### Use accurate document names

A pickup receipt proves the recorded handover. A processing record describes work performed. A Luma report summarizes platform evidence. An official recycling certificate, independently certified recycled content document, or carbon credit needs its own valid authority and process. Do not rename a Luma report as an official certificate.

**CURRENT RULE** Carbon credit work is later scope. Once a credit is issued, keep the original emissions factor. Later factor changes do not rewrite that issued credit.

### Manufacturing source history

The chat proposes following a material through fibre, yarn, fabric, and a garment. Keep the actual links and quantities, including blends, waste, and corrections. Bottle material used in clothing does not by itself prove recycled content in a new beverage bottle.

**Acceptance check** The same eligible quantity cannot support duplicate allocation within the same approved scheme. A reviewer cannot read unrelated business records. Estimated environmental benefit is visibly different from an issued credit.

Implementation decisions: D29. The selected answers are in section 23.

# 17 TASK 11 Partner services donations and rewards

*Source: Excel Stakeholders rows 23 27 to 36 and 44 to 51; Review Notes partner assumptions*

**PLATFORM RULE** A business may ask a partner for staffing, machinery, weighing equipment, transport, environmental engineering, accounting, or solar work. The source proposes a common enquiry, quotation, order, delivery, and payment process.

| Step | Person does this | Platform does this |
|---|---|---|
| 1 | Customer describes the service or equipment need. | Send the enquiry to selected eligible partners. |
| 2 | Partner submits a scope price and delivery plan. | Show included work, exclusions, charges, and validity. |
| 3 | Authorized customer accepts the quote. | Save the agreed scope, payment stages, and evidence required. |
| 4 | Partner records work delivery or installation. | Keep customer acceptance, service evidence, and outstanding work. |
| 5 | Customer or partner raises a service complaint. | Assign an owner and keep resolution and payment evidence. |

### Donations and worker programmes

The sheet proposes donations to NGOs, sponsored worker support, identity assistance, training, insurance links, and welfare enrolment. Each programme needs a named recipient, permission to share data, eligibility, funding, receipt, and complaint route. A referral is not a guarantee of enrolment, a loan, or an insurance benefit.

### Rewards and community campaigns

The sheet proposes points worth one rupee each, coupons, gold or jewellery redemption, community rewards, referrals, and campaign recognition. These remain commercial ideas. No signed merchant, available reward stock, funding model, expiry policy, or refund rule is established by the supplied files.

**Acceptance check** An unavailable partner does not appear as a confirmed booking. A reward is not issued or redeemed twice. Private identity or employment information is not sent to a partner without an approved basis and scope.

Implementation decisions: D07, D30, D31. The selected answers are in section 23.

# 18 Screens language and notices

*Source: Excel Stakeholders F M T and V; Chat screen descriptions; supplied UI instructions*

**PLATFORM RULE** Show the next useful action first. Use short text, clear photos where useful, and a simple progress view. Keep advanced business details available without making a household read them.

| Person | First view and main actions |
|---|---|
| Household | Recycle; own requests; next visit; receipt; payment record; complaint. |
| Worker | Assigned jobs; accept or decline; contact; weighing or handover evidence; own earnings. |
| Shop centre or yard | Incoming work; rates; received and available stock; sales; payments; day closing. |
| Plant or manufacturer | Orders; incoming material; quality holds; processing; outputs; source evidence. |
| Institution | Collection calendar; approved vendors; missed visits; stream totals and charges. |
| Reviewer or partner | Only the assigned review, service enquiry, or agreed report scope. |
| Luma operations | Exceptions; support; permitted verification; audit and operational reports. |

### Notices to describe

A request submission needs confirmation to its creator. An assignment needs a notice to the assigned person. Acceptance, rescheduling, cancellation, arrival, completion, failed payment, quality rejection, dispute, and permit expiry each need their own agreed recipient and next action.

**CURRENT RULE** Keep required consent and warnings visible. Preserve entered values between stages. Use existing translations and support right to left languages. Check light and dark screens on phone, tablet, and desktop. Household and account pages must not expose private information through optional analytics.

**Acceptance check** An empty list explains the next action. Missing service settings show unavailable. Long translations do not hide buttons. Notices on a locked phone use generic text rather than private material, identity, or payment details.

Implementation decisions: D32, D33. The selected answers are in section 23.

# 19 Records files privacy and correction

*Source: Excel inventory and evidence columns; Chat core business records; supplied project instructions*

**PLATFORM RULE** Keep these business records and their relationships. Their names describe business information, not a database design.

| Record group | Information to retain |
|---|---|
| Account business and site | Identity, contact, membership, approved access, location, activity, and relevant verification evidence. |
| Material and offer | Material name, form, grade, source, quantity basis, rate, validity, location, and terms. |
| Collection and work | Request, assignment, time, handover, actual weighing, photos, receipt, and worker fee. |
| Stock and processing | Lot, storage place, available quantity, reservations, input, outputs, additions, loss, quality, and correction history. |
| Order and transport | Accepted terms, seller, buyer, source and destination, load, vehicle, dispatch, receipt, and returns. |
| Money and service | Invoice, charges, payer, payee, payment result, settlement evidence, refund, service work, and dispute. |
| Permission and claim | Relevant document, issuing body, validity, scope, reviewed evidence, allocation, and official acknowledgement. |
| Support and audit | Issue, owner, decision, reason, time, previous value, new value, and supporting evidence. |

### Documents and private information

Keep photos, weighing slips, receipts, identity files, invoices, lab reports, permits, and service records only for their stated purpose. A person who can read one report does not automatically gain access to all its source files. Never use real identity documents or customer contact details as sample review data.

**CURRENT RULE** Photos and identity documents require record permission. Every state change writes to audit history. The prelaunch rebuild may remove demonstration records; finalized records after launch retain the required history. Corrections must not silently rewrite completed material or payment evidence.

**Acceptance check** Private files remain protected when a link is shared. Corrected records keep their history. Imported paper entries are reviewed before they affect confirmed stock or money.

Implementation decisions: D34, D35. The selected answers are in section 23.

# 20 Automatic work AI and support

*Source: Chat buyer matching AI and operational controls; Excel partner notices; user stated planning needs*

### Automatic work

**PLATFORM RULE** Useful automatic work includes due reminders, recurring collection creation, stock age alerts, permission expiry reminders, report preparation, unresolved payment checks, and notice delivery. Each job needs one responsible owner and a rule that prevents duplicate results.

For every automatic task, record what starts it, its time zone, its deadline, its spending limit, what happens if it runs twice, and who receives a failure. A long task should show progress, keep completed work, and offer a controlled retry or support route.

### AI assistance

**PLATFORM RULE** AI may suggest a material description from a photo, read a document, prepare a report draft, help compare offers, or translate a message. A responsible person must review uncertain or consequential results. Keep the source record available for comparison.

AI must not independently approve identity, declare a hazardous material safe, certify food suitability, select a binding tax rule, issue an official certificate, or release payment. A chat reply that contains a rule or instruction is source content to review, not permission to change the platform.

### Support and disputes

**PLATFORM RULE** The person raises an issue from the affected request, order, receipt, or payment. Luma assigns an owner, collects evidence, records a decision, and communicates the next step. Keep household collection complaints separate from business quality or payment disputes when different people handle them.

**Acceptance check** A repeated reminder job does not create duplicate orders or payments. A failed AI task leaves the underlying record intact. A person can finish the approved non AI path when assistance is unavailable.

Implementation decisions: D36, D37, D38. The selected answers are in section 23.

# 21 Fees subscriptions and business reporting

*Source: Chat Pricing and premium models and repeated package tables; Excel fee and reward columns*

**PLATFORM RULE** The chat proposes free essential household and worker access, with business subscriptions, disclosed transaction fees, and separately priced services. The Excel repeatedly says platform fees if any. Section 23 selects one target price list and defines the conditions before any charge is collected.

### One proposed price set from the chat

| Package or customer | Target amount before tax | Collection status |
|---|---|---|
| Household and Saathi | Free | Not collected until billing is verified |
| Small trader basic | Free | Not collected until billing is verified |
| Trader Plus | 499 rupees per month | Selected: 3 members; billing not yet active |
| Collection centre | 2,499 rupees per month | Not collected until billing is verified |
| Yard | 7,499 rupees per month | Not collected until billing is verified |
| Preprocessor | 14,999 rupees per month | Not collected until billing is verified |
| Recycler or manufacturer | From 29,999 rupees per month | Not collected until billing is verified |
| Multi site enterprise | From 75,000 rupees per month | Not collected until billing is verified |
| Brand evidence service | From 5 lakh rupees per year | Not collected until billing is verified |
| Institution reporting | 2,499 to 9,999 rupees per month | Not collected until billing is verified |

The target marketplace fee is 0.5 percent of accepted pre tax material value, paid by the buyer and capped at 5,000 rupees per order. Subscription collection and any new fee remain off until supported billing, merchant policy, and applicable tax configuration are verified. There is no automatic commission on external purchases recorded in the stock book or on internal processing.

Rules before charging

Disclose who pays and what they receive. Keep subscription, transaction, logistics, payment partner, lab, tax, and certification charges separate. The chat proposes one appropriate site package for a business with several roles rather than charging for each role. It also proposes no automatic marketplace fee for a legitimate external purchase merely recorded in the stock book.

**Reporting rule proposed** Show subscription income, transaction fees, service income, customer money passing through, and material trade value separately. Customer funds and total trade value are not automatically Luma income.

Implementation decisions: D39. The selected answers are in section 23.

# 22 Checks before approval and launch

*Source: Chat minimum acceptance tests; supplied project security and release instructions*

**PLATFORM RULE** Use these plain English checks to show whether the described tasks work. They are checks to perform, not test results.

| Area | What must be demonstrated |
|---|---|
| Sign in and sharing | Required checks cannot be skipped. Another business or a removed person cannot read a private record or file. |
| Household request | No weight estimate is required. Entered values remain after correction. Submitting twice makes one request. |
| Collection | Assigned people can record their work. One handover creates stock once. A failed visit remains distinct from completion. |
| Rates and stock | Old agreed rates remain unchanged. The same quantity cannot be reserved, sold, or consumed twice. |
| Processing | Inputs link to every output and residual. An unexplained difference prevents closure under the approved rule. |
| Quality and dispatch | Required checks and permission evidence apply before the relevant movement or use. |
| Payment and refund | Repeated notices do not pay twice. Unknown status stays unresolved. Refunds keep their evidence. |
| Claims and reports | A platform report is labelled accurately. An official claim uses the right evidence and does not duplicate allocated quantity. |
| Language and failure | Phone screens, keyboard use, translations, right to left text, and missing services remain usable. |
| Recovery | Restore agreed records and resolve interrupted work within the approved targets. |

**CURRENT RULE** Launch needs the existing checks and a check of the released service. The owner requires releases to the production service at luma.green. Release must follow the existing order and verification rules. A check on a test computer does not prove that a connected service works for real customers.

### What the coding agent receives

The coding agent receives this completed specification, the selected decisions in section 23, approved fictional examples, current project rules, and acceptance checks. Before building, it compares the plan with the existing platform. Section 23 resolves earlier source conflicts. A new material conflict must be raised with the owner.

Implementation decisions: D40. The selected answers are in section 23.

# 23 Selected implementation decisions

The following answers close the ordinary product gaps under the owner's instruction of 27 October 2026. They distinguish a selected product rule from an external fact. Required provider configuration and official permissions remain real service conditions. They are not filled with invented approvals.

### D01 Release scope

Build one connected platform for Bengaluru first. Core work covers household pickup and drop off, Saathi jobs, shop and centre stock, business buying and selling, processing, manufacturing source records, and scoped reports. The same business can buy, process, and sell. Specialist waste and partner services use the same record structure but remain unavailable for live execution until their required permissions and connections are verified. This is a service readiness condition, not an unanswered scope choice. The current release is web only. Expo iOS and Android work is paused; keep the existing work. Desktop software is excluded.

*Source basis: Excel rows 22 39 40 48; Chat phased scope and organic waste exclusions; current project rules.*

### D02 Success targets

Use a first acceptance workload of 50 simultaneous signed in people and 100 collection requests per day. Target ordinary screen responses within 2 seconds on a normal connection; show progress for longer work. Aim for at least 95 percent of accepted pickups completed in the agreed window. Support acknowledges ordinary issues within one business day; operations owns collection failures and accounts owns payment exceptions. These are build and pilot targets, not an advertised service guarantee.

*Source basis: Chat metrics and roadmap.*

### D03 Business membership

One person can belong to up to 10 workspaces. A business can own several workspaces or sites and hold several activities. Each workspace supports up to 100 members. A business owner grants membership, an authorized workspace admin manages member and viewer access, and only an owner changes ownership. A site activity never grants permission to perform regulated work by itself.

*Source basis: Excel G and H; Chat organisation model.*

### D04 Correction and deletion

Owners and authorized admins can remove unsubmitted drafts. Finalized receipts, movements, quality decisions, orders, and payment evidence are corrected by a recorded reversal or amendment with a reason. Never silently rewrite that history. During this prelaunch foundation rebuild only, the owner permits removal of obsolete demonstration records and schemas without migration or compatibility work. That temporary rebuild permission expires for future work once this foundation release is verified in Production. Security rules and the Expo pause remain active.

*Source basis: Excel D9 D15 and similar; current audit rules.*

### D05 Approval limits

Use owner, admin, member, and viewer as shared workspace access levels. Members record assigned operational work; viewers read only. Owners and admins manage teams and approved settings. Quality release and stock adjustments require a named responsible reviewer. Payment and refund operations retain the existing server checks and provider evidence; no person can mark a gateway payment successful. Do not create a second payment approval system merely because a spreadsheet names an accounts role.

*Source basis: Chat internal permission controls.*

### D06 Household sharing

A household account is private by default. Share a specific collection only through an explicit verified adult invitation. A helper may view and record assigned handover work but cannot change rates, payment destination, or other family records. Sharing ends on revocation or completion of the assigned task. Do not create accounts for children in this version. Keep a clear leave or revoke action.

*Source basis: Excel G2 H2 and rows 3 to 8.*

### D07 Partner scope

Core trading and operating accounts use Luma workspaces. Equipment, staffing, transport, environmental engineering, accounting, solar, scale, voice, society app, and welfare services are partner enquiries with scoped service records in this version. They are not independent copies of the trade system. An actual confirmed booking requires a configured partner. Unconfigured partners show unavailable. Rewards, lending, and official credit market execution remain unavailable while their required providers and permissions are absent.

*Source basis: Review Notes assumptions 6 7 and 9.*

### D08 Business activation

Retain current valid business activation without a blanket 12 to 24 hour manual queue. Keep the current Surepass business registration check and explicit name and address confirmation. Retain the newer optional business PAN to GST discovery from current main, excluding household, Saathi, and admin accounts. Tax registration is separate from operating permission. A required facility permission review gates the affected activity, not ordinary sign in.

*Source basis: Excel H26 V26; current application rules.*

### D09 Lost and shared phones

Keep verified phone sign in and all required authenticator checks. Shared phones do not mean shared sessions: show the active account and provide sign out. Lost phone recovery uses the existing verified recovery path and required second factors; support cannot bypass them. A departing worker loses membership immediately and the owner reassigns unfinished work. Phone binding preserves the existing identity and its security factors.

*Source basis: Excel T and linked users; current authentication rules.*

### D10 Suspension

Suspension immediately blocks new orders, new work acceptance, and new operational changes by the suspended party. Keep existing receipts and permitted read access needed for resolution. Platform operations and the authorized counterparty can finish a dispute without granting the suspended party new rights. Only the platform admin can restore a platform suspension; an owner manages workspace membership separately.

*Source basis: Excel B2 and row 26.*

### D11 Assigning collection

A household request is offered to eligible active shops that serve its location and have capacity for the selected window. The first permitted acceptance reserves the request once. Do not auto assign an ineligible shop. If no shop accepts within two hours, show still unassigned and offer another time or cancellation. Operations can reassign only to an eligible accepting shop and records the reason.

*Source basis: Excel I2 I9 and M9; I9 is incomplete.*

### D12 Times and cancellation

Offer two hour pickup windows between 9 AM and 6 PM India time for the next seven days, limited by each shop availability. Drop off uses only a business published site and open hours. The household can cancel or change the time before collection starts. After the worker starts travel, a change becomes a reschedule request for the operator. A missed visit keeps its reason and permits a new booking; do not claim completion.

*Source basis: Excel M2 M9; household task.*

### D13 Household estimates

Do not require household material names, weights, or a stock estimate. Photos and the collection details are sufficient before handover. The shop records actual materials and weights. Leave household stock valuation out of the first version. Any displayed material rate is an offer or dated reference, never a promised final total before weighing.

*Source basis: Excel N2 to O8; current household rules.*

### D14 Weighing and worker pay

The shop records the material lines, measured quantities and agreed rates. An assigned worker records a collected total and evidence; the employer compares it with the completed shop receipt. The handover never moves stock or money again. The household sees and confirms or disputes the receipt before acknowledging payment. A rejected line is excluded from accepted quantity and kept in the history. Worker fees are a fixed agreed job fee for this version, disclosed before acceptance. There is no default failed visit charge, hidden deduction, or platform wage guarantee. The assigning business pays the worker and records the due date and acknowledgement.

*Source basis: Excel K14 P14 and field worker rows.*

### D15 Location tracking

Use explicit foreground location permission only for the active task. The assigned operator and requester see only the location information needed for that task. Stop sharing on completion, cancellation, sign out, or revocation. Manual address and status entry remain available. Do not use a 30 metre trigger, photo location, or a map position as proof of handover or payment. Do not add background employee tracking in this version.

*Source basis: Excel I2 I9 U11 and row 36; current permission rules.*

### D16 Minimum price

Businesses set their own offers and contract rates. Do not enforce one compulsory platform minimum rate. Keep dated indicative market information separate from binding offers. Historical prototype floor settings do not authorize a new trade restriction. An accepted order keeps its rate and terms after the public reference changes.

*Source basis: Excel J and K; Chat corrections and pricing section.*

### D17 Market rate board

Show a business own current offer with its validity. Public reference information must identify its source and date; show unavailable when missing. Treat a business rate older than 24 hours as stale unless it has a longer explicit contract validity. Any future completed trade benchmark uses compatible grades and geography, excludes demo and self trades, and requires at least five independent contributors. It does not replace the agreed price.

*Source basis: Excel J2 unfinished; Chat reference price board.*

### D18 Stock adjustments

Use the append only movement record as stock authority. A correction needs a reason and responsible owner or admin; affected completed trades are resolved by amendments rather than editing old quantities. Reserve and consume exact grams once. Use exact recorded acquisition and processing costs for value, with a separately labelled current offer value. Low stock alerts are opt in business settings rather than guessed industry thresholds.

*Source basis: Excel N and O; Chat stock controls.*

### D19 Matching and quotes

Required matching checks are active permitted account, accepted material and grade, service area, available quantity or capacity, required valid evidence, and compatible handling. Rank eligible results by delivered net value when prices exist, then distance and earliest availability. A missing price is request a quote, not zero. Use direct quotes and accepted orders first; auctions and tender management are outside this version.

*Source basis: Chat eligibility and ranking; Excel M19 M38.*

### D20 Order types

Use one complete accepted quantity per business order in this version. Retain the existing full collection, dispatch, receipt, cancellation, and refund policy. Do not add part deliveries, part acceptance, credit lines, instalments, or advances. Longer supply relationships can create repeated complete orders with their own agreed terms.

*Source basis: Chat buying and procurement examples; current bounded payment process.*

### D21 Money movement

A shop pays a household directly and Luma records the separate payment. Keep Cashfree as the selected business payment provider. Live business collection and split settlement require its configured approved policy and provider acceptance. Do not simulate escrow or add a manual paid override. Rewards, gold, lending, and official certificate trading do not gain a money movement path by being listed as services.

*Source basis: Excel P to S; Review Notes 2; Chat payment alternatives; current rules.*

### D22 Payment and disputes

Record household cash or UPI/bank payment with payer, recipient, exact amount, date, method, and household acknowledgement or a recorded dispute. Business charges are disclosed before checkout; the buyer pays the disclosed platform fee. Preserve current Cashfree payment, cancellation, and full remaining refund guards. Unknown payment remains pending until verified. Operations acknowledges disputes within one business day; keep all weighing and provider evidence. A refund does not itself restore stock.

*Source basis: Excel payment columns; Chat settlement sections.*

### D23 Tax and movement documents

Store transaction facts and the reviewed rule version with each applicable document. Generate only supported commercial receipts and evidence drafts until an applicable official tax or movement rule is verified. Never infer a universal tax rate, consent duration, form number, or filing permission from the chat. Missing required legal evidence blocks that particular movement. This closes the product behavior without inventing legal facts.

*Source basis: Chat Parts A and F and later consignment correction.*

### D24 Processing rules

Provide source labelled reference definitions for sorting, baling, washing, shredding, pellet, fibre, yarn, fabric, and garment records. Each business supplies its actual input, output, additions, residuals, measured loss, test method, and contract specification. A run closes only when quantities reconcile in exact grams and all required results are present. No universal yield or tolerance is invented. Draft reference definitions do not claim approved product quality.

*Source basis: Excel row 17; Chat PET and processing examples.*

### D25 Quality differences and returns

Keep seller and receiver weights and test evidence separately. The accepted order defines the quantity basis and quality tests; no automatic blanket percentage tolerance or double deduction. A mismatch creates a hold and dispute. The responsible buyer reviewer decides accept or reject under the agreed complete quantity policy. A return requires an eligible destination, recorded freight payer, approved disposition, and reconciliation of stock, money, and claims.

*Source basis: Chat quality and 10,000 versus 9,820 kilogram example.*

### D26 Recurring and wet waste services

Support a repeating collection schedule as separate dated requests. Changing or ending the schedule affects future requests only. Keep a missed visit and its reason. Wet waste uses a distinct service enquiry and processing record path and cannot enter ordinary dry recyclable trade. Specialist handling remains unavailable without its verified receiver and rules. Holidays are defined by the business availability calendar; do not create visits outside it.

*Source basis: Excel rows 22 40 48; Chat organic waste exclusions.*

### D27 Public programme agreements

Use scoped programme workspaces and aggregate area reports. A municipal or rural partner sees only an explicitly shared programme, area, period, and stream. Do not assume an agreement with a named public body. Record coverage, trips, centre handovers, service references, and group income evidence. Public contract payments and welfare awards remain partner decisions and are not fabricated by Luma.

*Source basis: Excel rows 24 40 41 42.*

### D28 Specialist material scope

Keep separate material families and reference sources for ordinary dry recyclables, textiles, electronics, batteries, tyres, used cooking oil, lubricating oil, industrial residues, and end of life vehicles. Commercial category is distinct from legal waste classification. Do not adopt either 359 or 419 as a verified official list. Unknown or restricted streams require a verified permitted route and cannot be booked as ordinary household scrap. Biomedical waste and controlled drug destruction are excluded from this version.

*Source basis: Chat industry counts and unprovided referenced attachments; Excel 39.*

### D29 Environmental reports and certificates

Provide collection receipts, processing records, source history reports, and clearly labelled evidence packs. Record valid official references when supplied by an authorized issuer. Do not issue official EPR certificates, certified recycled content documents, vehicle scrapping certificates, or carbon credits as ordinary Luma reports. Track allocation within each approved scheme and withdraw affected evidence after an invalid source or return. Credit market execution is unavailable until its own provider conditions pass.

*Source basis: Excel rows 21 30 31 43; Chat claims; current carbon scope.*

### D30 Rewards and donations

Donation and reward records can describe an enquiry or programme, but no balance with monetary value is promised or issued in this version. Do not adopt one point equals one rupee, coupons, or gold redemption without funded merchant terms. The interface shows unavailable for redemption. Direct donations require a named approved recipient and receipt; Luma does not quietly divert household proceeds.

*Source basis: Excel P2 to S2 and rows 23 45 50.*

### D31 Finance welfare and attendance

Loans, insurance, welfare enrolment, staffing compliance, and attendance are scoped partner referrals or service records. The provider decides eligibility and handles its regulated service. Share only the specifically authorized information. Luma support tracks referral complaints without promising a loan, cover, government benefit, or employee monitoring capability.

*Source basis: Excel rows 25 27 36 46.*

### D32 Notice rules

The account inbox is the record of user notices. Send existing configured SMS for essential collection and account events, and email for its existing invitation or recovery purposes. Optional push requires explicit device permission and generic lock screen text. Deduplicate each event. Non urgent reminders run from 8 AM to 8 PM India time. Payment and sign in security messages are immediate. WhatsApp and voice show unavailable until configured, without blocking the app path.

*Source basis: Excel M; Chat notices.*

### D33 Voice and language support

Keep all 33 registered languages and script aware fonts. Verify English, Kannada, Arabic and Urdu layouts as representative coverage, then preserve key parity across all. Use icons with accessible text; permit assisted entry with the account holder consent. Phone browser use remains in scope. Expo work is paused. Do not add a separate voice or WhatsApp product to finish this version.

*Source basis: Excel T and row 51; current translation rules.*

### D34 Retention and access withdrawal

Private data follows record access. Revoking sharing removes later access immediately. Keep finalized operational and payment evidence for the published retention policy; the release must not publish a fabricated statutory period. Unsubmitted abandoned drafts expire after 30 days unless linked to an open issue. Pending uploads expire after 24 hours. Account removal removes access and optional profile data while preserving required anonymized or restricted audit evidence under the reviewed retention policy.

*Source basis: Chat privacy and audit; current record preservation.*

### D35 Uploads and paper entry

Accept only supported image or document formats, validate actual content, and enforce existing upload limits. Use private R2 for the configured supported file scope and retain working Convex storage for existing identity or unconfigured uploads until the common private upload path is ready. Never make a bucket public. A wrong or expired required document needs replacement and keeps the action blocked. Paper logbook extraction is an AI assisted draft that a person confirms before it affects stock or money.

*Source basis: Excel row 15 and evidence columns.*

### D36 Automatic tasks

Use Convex scheduling for short transactional follow up and local cleanup. Use Trigger.dev for bounded long reports, document extraction, and AI work. Each job has one scheduler owner, a unique operation key, visible progress, a deadline, an execution limit, and a recorded failure. No automatic retry for a paid GST or OTP operation. Internal work can retry up to three times with delay when it is safe to repeat. India time owns business schedules; stored timestamps use a consistent universal time.

*Source basis: Chat expiry and daily jobs; requested automatic work.*

### D37 AI assistance

AI can suggest a material description, extract a document, translate a draft, or summarize source records. It cannot grant permissions, set binding tax, approve a hazardous route, issue official evidence, change confirmed quantities, or pay. A person confirms any proposed saved change. Keep existing server quotas and configured model budget. Without an approved configured budget, the optional AI service is unavailable and the manual path remains usable.

*Source basis: Chat AI limits; owner planning request.*

### D38 Support and recovery

Provide one support record linked to the request, stock record, order, service, or payment. Operations handles collection and access issues; accounts handles money; a qualified reviewer handles specialist evidence. Acknowledge within one business day and escalate unresolved payment or safety issues immediately to the responsible operator. Target restoring records to the last successful daily backup within one business day. These are internal acceptance targets until a published support commitment is approved.

*Source basis: Excel V26; Chat resilience and acceptance checks.*

### D39 Prices and fees

Use one site subscription per business activity bundle, not separate charges for buyer and seller roles. Keep essential household and worker access free. The target monthly business tiers are Trader Plus 499 rupees with 3 members, Centre 2,499 with 5, Yard 7,499 with 10, Processor 14,999 with 15, Recycler/Manufacturer from 29,999 with 25, Enterprise from 75,000, and institution reporting 2,499 to 9,999 by agreed scope. Live subscription collection stays off until supported billing and applicable tax configuration are verified. A future disclosed marketplace fee is 0.5 percent of accepted pre tax material value, paid by the buyer and capped at 5,000 rupees per order; external records and internal processing attract no automatic commission. No charge is collected merely because this target price list exists.

*Source basis: Chat multiple price tables; Excel platform fees if any.*

### D40 Final business approval

The platform owner is the final business and release approver. The implementation team records evidence for each task and configured provider. Ship only after required checks and current Production verification. Expo device and store acceptance are deferred while app work is paused. Use internal test participants and fictional isolated fixtures before real transactions. If a release fails, stop affected writes and restore the previous working release; preserve the new evidence for repair.

*Source basis: Chat launch gates; current release rules.*

# 24 Useful paid services

Checked 27 October 2026. Prices below are in US dollars before tax and usage charges. They are purchase recommendations. No plan has been bought or changed.

### Vercel Pro

20 per month. Web hosting and commercial team controls.

https://vercel.com/pricing

### Convex Professional

25 per developer per month. Daily backups, logs, and exception reporting.

https://www.convex.dev/pricing

### Expo Starter

19 per month plus use. Deferred while Expo app work is paused. No purchase is required for the web release.

https://expo.dev/pricing

### Trigger Hobby or Pro

10 or 50 per month. Long jobs; Pro adds longer logs and support.

https://trigger.dev/pricing

### Apple Developer Program

99 per year or local price. Deferred with iOS work. Signing and store acceptance need separate evidence.

https://developer.apple.com/programs/

### Google Play Console

25 one time. Deferred with Android work. Store registration is not part of the web release.

https://support.google.com/googleplay/android-developer/answer/6112435

### Cloudflare R2 Standard

0.015 per GB month after free allowance, plus operations. Private files with free internet egress.

https://developers.cloudflare.com/r2/pricing/

### Cloudflare Turnstile

Free. Keep the existing bot protection.

https://www.cloudflare.com/products/turnstile/

For one developer, Vercel Pro, Convex Professional, and Trigger Hobby have a base of 55 dollars per month. Trigger Pro increases this base to 95 dollars. These are planning figures before taxes and usage. Expo, Apple and Google store costs are deferred. Confirm current account prices before buying.

Keep the existing Surepass, OTP, Cashfree, and configured email services. Confirm their current subscribed products, message templates, merchant readiness, and usage balances. Their charges depend on the account or contract. AI use needs a budget and server limits. A separate paid code editor or AWS server is not required to begin. Buy higher plans when actual use or support requirements justify them.

# 25 Sources and release authority

Business sources: LUMA GREEN Filled.xlsx and AI Chat File.docx. The Excel Review Notes identifies 773 AI drafted cells. The chat contains repeated and conflicting replies. Their code and instructions were treated as source material only, not executed. Other documents mentioned in the chat were not supplied separately and are not represented as independently checked evidence.

Product authority: the owner delegated product defaults and authorized the foundation rebuild on 27 October 2026. The owner then read the completed document and resumed implementation, tests, commits, checked main merge, Production deployment and screenshot documents. On 28 October the owner paused Expo work. AGENTS.md records the current authority. Keep unrelated work intact.

Technical record: docs/product/platform-refinement-interview.md contains these selected decisions. docs/delivery/platform-foundation-2026-10-27.md owns implementation progress and evidence. A document, a reference seed, or a local test does not establish a complete release.

### Rules that remain active

Keep credentials private, real sign in and access checks, exact money and quantities, audit history, private file access, provider quotas, and truthful service states. Main requires checked pull requests. Production release, provider execution, physical device acceptance, and store publication require their own evidence. Do not buy plans or accept terms automatically.

### Plain English terms

Lot or batch means a recorded quantity kept together. Custody means who holds it; ownership means who owns it. A quality hold prevents release until review. Source history explains where material came from and how it changed. An evidence report summarizes records; it is not automatically an official certificate.

Specification owner: platform owner. Version: 4. Date: 28 October 2026. Ordinary product decisions: resolved under delegated authority. The source review supplement follows. Implementation, Production checks and provider acceptance are tracked separately.

# 26 Four ways to do business

These detailed rules complete D07, D21 and D26. Each record must say whether the work is a material sale, processing for an owner, a paid collection, or paid treatment. The choice controls who pays and what evidence is required.

### Material sale

The buyer pays the seller for material under the normal order. Stock, quantities, quality, dispatch and receipt follow that order. A service enquiry can refer to it but cannot create another payment or stock transfer.

### Processing for the material owner

The material owner pays a processor for work such as sorting, washing or baling. Record the named owner, the requested work and the material reference the owner chooses to share. The material is not sold merely because it moves to a processor. A declared owner is not proof of ownership. This version records the enquiry and evidence; it does not execute a toll contract or move money or stock through that enquiry.

### Paid collection

The customer pays the collection provider for a service. Record the work, parties and requested terms. Keep the service fee separate from any material purchase. The enquiry cannot pretend that an outside provider has confirmed a visit or accepted a price.

### Paid treatment

The waste generator pays a qualified handler for treatment or disposal. Record it as a service request. Do not enter a negative scrap sale price. Wet waste has its own service kind. A restricted stream cannot use the ordinary dry scrap booking or trading route. Live treatment remains unavailable until the receiver, permissions and operating route are verified.

### One clear service state

A person can save a private draft, send an enquiry to a chosen approved recipient, edit it, close it or withdraw recipient access. A retry of the same creation request returns the same record. A changed request cannot reuse its reference. The record says that its contents are reported and unverified. No enquiry alone confirms delivery, payment, eligibility or legal approval.

# 27 Private services and programme reports

These rules complete D07, D27, D30 and D31. Use the same private service records for equipment, staff, transport, scales, accounting, environment, solar, society support, finance, insurance, welfare, attendance, donations, rewards, voice and other referrals. Wet waste remains a separate kind.

### Choose what is shared

Only the creator, the chosen currently approved recipient and an authorized platform admin can read an enquiry. Other members of the same business do not gain access merely by membership. The creator chooses whether to send it. Removing a recipient or suspending that recipient removes later access. A material reference shares only the chosen material code and recorded quantity. It does not open the underlying lot, its files or its source history.

### Record a programme

The programme owner records its name, area, date period and material stream. Record coverage, trip and centre handover counts, measured quantity, reported group income and a private evidence reference. Keep quantities in exact grams and money in exact paise. Mark the figures reported and unverified. The record does not establish a public agreement, audited income or government award.

### Share an aggregate report

The owner selects one approved recipient, a purpose and an expiry of no more than 90 days. The report is a fixed copy of the area, period, stream and totals. It excludes people, contacts, enquiry text and private evidence. Changes to the programme require a new report. Revocation ends later access to that report. It does not silently change the fixed copy.

### Keep promises within the service limits

A finance or insurance referral promises no loan or cover. Welfare and attendance records promise no benefit or background tracking. A donation enquiry promises no transfer. A reward enquiry creates no spendable points, coupons or gold. Actual donations need a separately approved recipient, transfer and receipt path. Configuring a provider is a separate step from making its reference label available.

### Support and retention

A permitted participant can link a private complaint to a service or programme. Record the responsible team and updates. Unsubmitted drafts expire after 30 days of inactivity. An open linked issue keeps its draft. A programme with a published aggregate report keeps the source evidence, even after report access is revoked.

# 28 Returns claims and buyer comparison

These detailed rules complete D25 and D29. Keep the original sale and its accepted quantities unchanged. A return is a new linked record with its own physical movements and financial status.

### Agree the full return

Either active order party can propose a return after the buyer has physically received the material. This version supports the full accepted quantity. Record a unique reference, reason, the original seller as destination and which party pays freight. The original seller must still be active and eligible. Another destination or a partial return is unavailable in this version. The other party must approve or reject.

### Keep stock and money separate

Approval holds the returned quantity so that it cannot be sold again. The buyer records dispatch once. The seller records exact receipt once. Repeating an identical action cannot move stock twice. Restore the seller recorded dispatch cost only when it is known; otherwise show unknown. A confirmed refund cannot restore stock by itself.

Financial reconciliation requires the live collection, confirmed full refund and provider settlement reversal evidence. Without them, show pending. No user or admin control can invent refunded or reversed payment status. Physical return, refund and settlement remain separate visible facts.

### Withdraw affected claims

An organization can report an allocation against an eligible owned received purchase, under one named scheme and an external issuer reference. Current accepted quality evidence must remain valid. The combined allocation for that source and scheme cannot exceed the received quantity. Label the declaration unverified. Luma does not issue an official certificate or credit.

When a return is approved, withdraw the affected declarations in the same saved change. An invalid source must not remain presented as an active claim. Preserve earlier declarations and reasons in the audit history. Corrections create new evidence; they do not rewrite past references.

### Compare offers with known facts

Compare eligible offers for the same material and matching declared grade or specification. Show lower known material plus freight cost first, then known shorter straight line distance and earlier recorded availability. Missing freight means quote required, not free delivery. Keep the offer expiry visible. Recheck available stock before request and acceptance. Accepted orders keep their agreed terms. Freight is agreed and paid separately from the material checkout.

### Record actual processing cost

Record a known acquisition or processing cost with its evidence reference. An empty cost means unknown; explicit zero means a recorded zero. Consume input cost in exact proportion to used grams and keep the remainder. Allocate input plus processing cost across measured outputs, including residuals, without losing paise. Surviving outputs absorb process loss cost. If any required input or processing cost is unknown, the output cost stays unknown. Production reads this cost once. A manufacturer stock intake separately records the total cost allocated to that exact intake; never add the same processing cost again. Cost evidence does not prove ownership, quality or market value.

# 29 Location monitoring and release evidence

### Ask before finding an address

Manual address entry always remains available. A person can choose a location, then separately agree to send that point to Google Maps and select Find address. A verified phone session owns the request. Limit it to ten attempts per account per India day, with spacing and duplicate protection. Failed attempts count. Do not retry a paid request automatically.

Show a returned address with Google Maps attribution. Never replace entered text automatically. Keep the suggestion only in the current screen. Account, point or address changes clear it. Saving Google address text stays disabled until the owner completes the provider policy review. When saving is allowed, the person must check the address and explicitly choose to use it. A point does not prove the correct building.

### Measure public use with consent

PostHog and Google Analytics are optional. They start only after the visitor allows analytics and the owner configures the service. Measure only the selected public information pages. Keep login, forms, tracking, workspaces and admin outside analytics. Do not capture form contents, contacts, identities, booking tokens or private record contents. Keep automatic capture and session replay off. The visitor can turn analytics off later.

Sentry is a separate optional error service. Enable it explicitly and remove identifying details from error context. Set account budgets, retention, alerts and named owners before activation. Provider settings and a local test do not prove that a live event or alert arrived.

### Separate definitions from example records

Reference initialization supplies 51 material labels and 32 draft definitions: 21 PET processing states and 11 specialist references across eight categories. New PET labels stay inactive until review. Specialist references stay inactive and their routes unavailable; an active flag cannot authorize their trade. Service kinds have a separate initializer. These writes create no live rates, stock, customer identity, payment, permit, certificate or provider approval. Fictional participants and sample journeys remain isolated test data.

### Check the full source and the release

The source register accounts for 84 capability groups from the workbook and chat. Ordinary product rules are answered. Seven groups retain proof, operational, outside-readiness or optional-work limits. All 28 participant keys, 21 PET states and 11 specialist references have offline coverage. This does not prove every browser journey, factory recipe or provider call. Release still needs required checks, a checked main merge and Production verification.

Expo iOS and Android work is paused. Its existing changes are preserved outside the web release, and the queued Android build was cancelled. Signing, device tests and app publication are deferred. Label example screens and actual Production captures accurately. Recovery drills, measured capacity, provider acceptance and legal approvals need their own evidence before any related promise is published.

# 30 Required completion checks

The product rules below are decided. The source register separates code ownership from acceptance evidence. Remaining checks concern release proof, outside services and optional paper extraction. The delivery record tracks final checks and Production verification.

### Collections and worker handover

Check collection acceptance against location, capacity and the two hour unassigned limit. An employer binds an assigned pickup job to its own visit. The worker records collected grams and an evidence reference. The employer derives received grams from the completed shop receipt. Keep the difference and require a reason when it is not zero. A stale assignment or schedule cannot authorize another visit. This adds no stock, payment or wage change and gives no household contact access. Independent worker owned trading and day bag inventory are outside this version. Institution reports need a permitted, scoped journey.

### Stock and actual costs

Record one fixed closing for the current India date after reconciling stock and movements at recording time. Keep exact grams and recorded cost; unknown stays unknown. Later movements keep their own history. Low stock notices are off until an operator chooses a material and threshold. One downward crossing creates one inbox alert; recovery allows a later alert. An owner or admin can correct stock with a reason and evidence, while preserving reservations. Corrections never rewrite past receipts, orders or payments. Combined release acceptance remains separate.

### Matching and specialist streams

Check delivered value against current area, stock, grade and required evidence. Specialist references identify separate categories, not verified legal classifications. Their ordinary sale and stock intake routes remain blocked. Live specialist handling needs its own verified receiver and route. Draft PET tests do not validate a factory recipe.

### Account removal and optional paper entry

Removal needs explicit confirmation after a full sign in within five minutes, including any required authenticator. It holds while the person remains the site owner or last business owner, or has an open visit, assigned job or unfinished handover. Held means not closed: resolve the work or transfer ownership first. Personal stakeholder registration history does not block closure. Eligible closure ends private access, sessions, devices, memberships, personal grants and keys at once. Clear optional profile settings; finish cleanup in batches of at most 100 records. Keep necessary history under restricted admin review, without inventing a legal retention period. Closing never refunds money, moves stock or confirms handover. A new account using the same phone gets no old records or grants and cannot reset the paid business check limit for that day in India. The configured admin needs a separate replacement path. Optional paper extraction and confirmation are not implemented; manual entry works. It needs its own draft and confirmation workflow as well as approved AI setup.

### Screens recovery and capacity

Check current role screens, phone and desktop sizes, themes and representative languages. Label fictional examples. Run the recovery drill and measure the pilot target of 50 simultaneous signed in people and 100 requests per day. A successful build or backup setting does not prove these results.

### Outside service acceptance

Keep outside services unavailable where required provider or policy conditions are missing. This includes payments, official filings and certificates, specialist treatment, funded benefits, optional messaging, AI, long reports and address lookup. Record the setup owner and next step. Do not invent approval. Expo and desktop remain deferred.